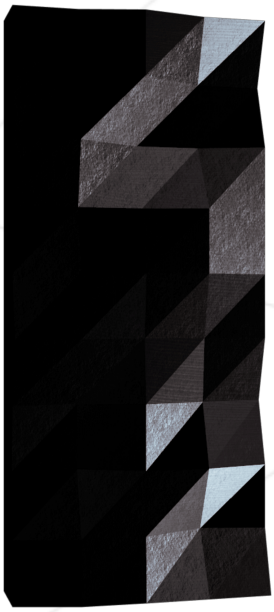

Active Discovery - Formation ActiveDirectory

Initiation

Summary: This project introduces you to the basics of system administration under the Microsoft Server operating system.

Version: 1.1

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Chapter I

Preamble

This subject is the production of a partnership between 42 and Microsoft Corporation.

Microsoft Corporation is an American multinational technology corporation headquartered in Redmond, Washington.Microsoft's best-known software products are the Windows line of operating systems, the Microsoft Office suite, and the Internet Explorer and Edge web browsers.

Through this partnership, we aim to provide you with a unique opportunity to simplify your journey towards obtaining certifications offered by Microsoft. These certifications hold significant value and recognition in the industry, enabling you to enhance your professional profile and unlock exciting career prospects in the field of security.

To access the Microsoft certification programs and explore the wide range of certificates available, you can visit the following link: Microsoft Security Certification. This comprehensive platform offers extensive training resources and examinations covering various security-related topics.

By successfully completing the projects and earning Microsoft certifications, you will demonstrate your proficiency and expertise in security practices. These certifications serve as a testament to your skills and can open doors to exciting career opportunities.

Figure I.1: https://www.microsoft.com/

Chapter II

AI Instructions

Context

AI is now a powerful coding partner — alongside your peers — for tackling large and demanding projects. You will guide it through both technical and non-technical aspects of your work.

AI tools can boost your efficiency and improve the quality of your output, but you should be able to dive deep into any part of the project without relying on them.

Your AI partner supports you, but you remain fully responsible for making informed technical decisions and to clearly explain and defend them.

Main message

- Strive for a mature and responsible use of AI.
- Never let AI take responsibility for decisions — especially when it lacks awareness of your goals, constraints, or team dynamics.
- Maintain creativity, innovation, and human oversight through active collaboration with your peers. AI is trained on existing data and rarely generates truly new ideas.
- Stay informed about emerging trends and be ready to adapt to new concepts and technologies.

Learner rules:

- Maintain intellectual leadership over your projects and make your own informed decisions.
- Prioritise the collective intelligence of your team and peers.
- Actively stay informed about the ongoing evolution of AI technologies.

Phase outcomes:

- AI engineering skills.
- Increased efficiency.
- Greater reliability and quality.
- A pioneering mindset.

Comments and examples:

- Your peers can identify trade-offs, question assumptions, and help you improve. The first answer from an AI might not be the best — it may lack efficiency, security, or real added value. Now more than ever, you should rely on your peers.
- AI can make you faster, but your peers make you better. Collaboration, discussion, and mutual challenge are key to success.
- Be transparent about how AI was used in your projects, and clearly identify what was generated by AI tools.

✓ Good practice:

> I asked AI to help generate unit tests for my API. I reviewed them with my teammate, and we adjusted them for edge cases. It saved time, and we both learned something new.

✗ Bad practice:

> I had AI generate the entire architecture of my project. It "works," but when I'm asked to explain design decisions during the peer review or in front from of a customer, I cannot. I lose credibility and I fail.

Chapter III

Introduction

In partnership with Microsoft

Active Directory (AD) is Microsoft's implementation of LDAP directory services for Windows operating systems.

The primary objective of Active Directory is to offer centralized identification and authentication services to a network of computers using Windows, macOS, and Linux. It also facilitates the administration of policies, enforcement of policies, and installation of critical updates by administrators.

Active Directory provides a comprehensive listing of managed network elements such as user accounts, servers, workstations, shared folders, printers, and more. It enables users to easily locate shared resources, while administrators can exercise control over their usage through distribution, replication, partitioning, and secure access to the listed resources.

In this project, you are introduced to the functionality and features of Active Directory.

You're provided with a helper pdf, foundable inside the attachments of the subject.
It will show you how to access the VMs created for you to complete this subject, how to lock your VMs, etc.

Chapter IV

Mandatory part

IV.1 Presentation

In this project, you will install and configure the computer infrastructure for "Domolia," a company that specializes in selling home automation tools.

The company is divided into two buildings:

- A workshop, managed by a supervisor, with several workers.
- An office, where the company manager is located.

The company requires two Microsoft Server installations, one in each building, along with a Windows workstation that will connect to the respective building's server.

Ensure that you use a long-term support (LTS) version of Microsoft.

Your objective is to set up virtual machines that closely resemble the desired network organization.

Remember: You are a team of two students, and this project likely requires more than just the power of friendship!

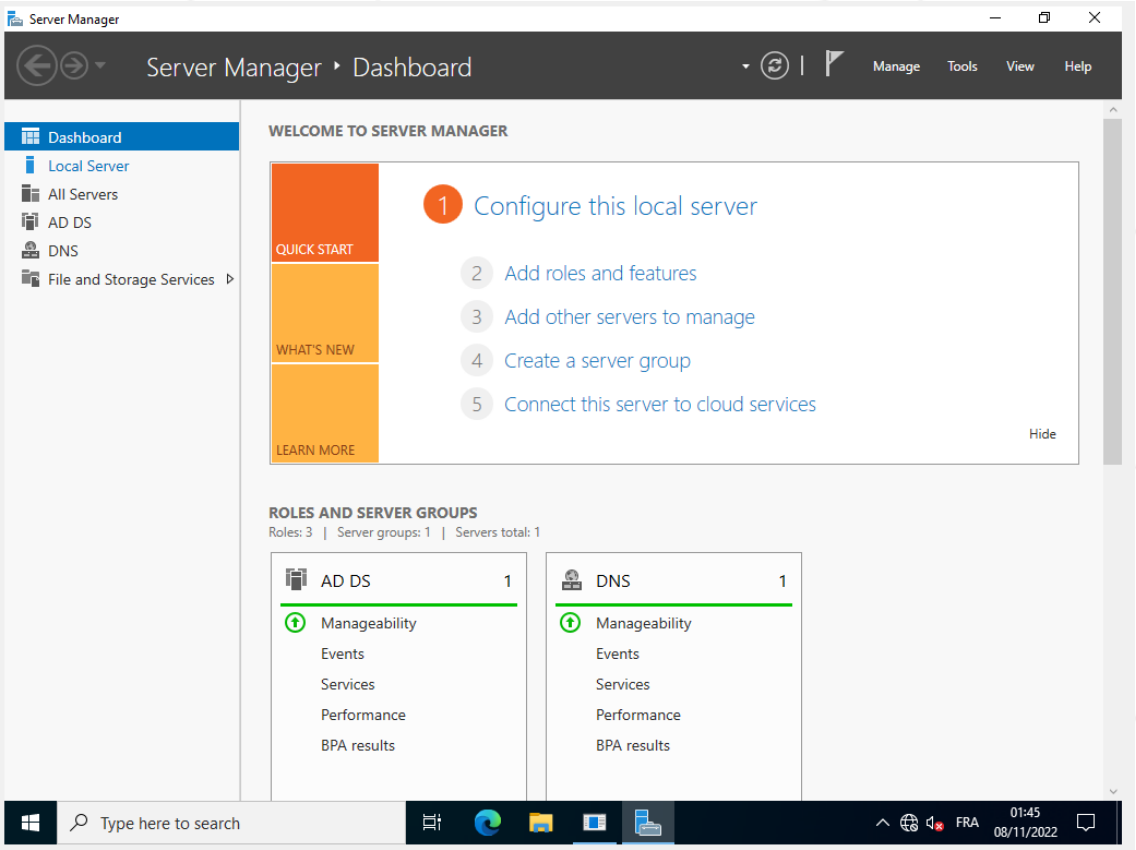
You're provided with a helper pdf, foundable inside the attachments of the subject.
It will show you how to access the VMs created for you to complete this subject, how to lock your VMs, etc.

IV.2 Forest creation

In this part, you have to install and configure the server located in the administration building.

The server must have the Active Directory Domain Controller service installed and configured. You need to add any necessary modules you think are required for this service.

The server must have a suitable name that allows easy identification on the network.
You should create a forest on the server and choose an intelligent name for it.



When you finish this part of the project, you should have something like this on your screen.

IV.3 Domain controller configuration

In this part, you need to connect the second server to the first one in order to enable it to join the existing forest.

This server will be responsible for hosting its own domain controller.

Similar to the previous server, you should choose an appropriate name for this server.

IV.4 Resources creation

In this part, you are required to create three different folders where users can store important files.

- Create a folder on the Administration server specifically for storing administrative files.
- Create another folder on the Administration server for storing generic data and files.
- Lastly, create a folder on the Workshop server dedicated to storing working project files.

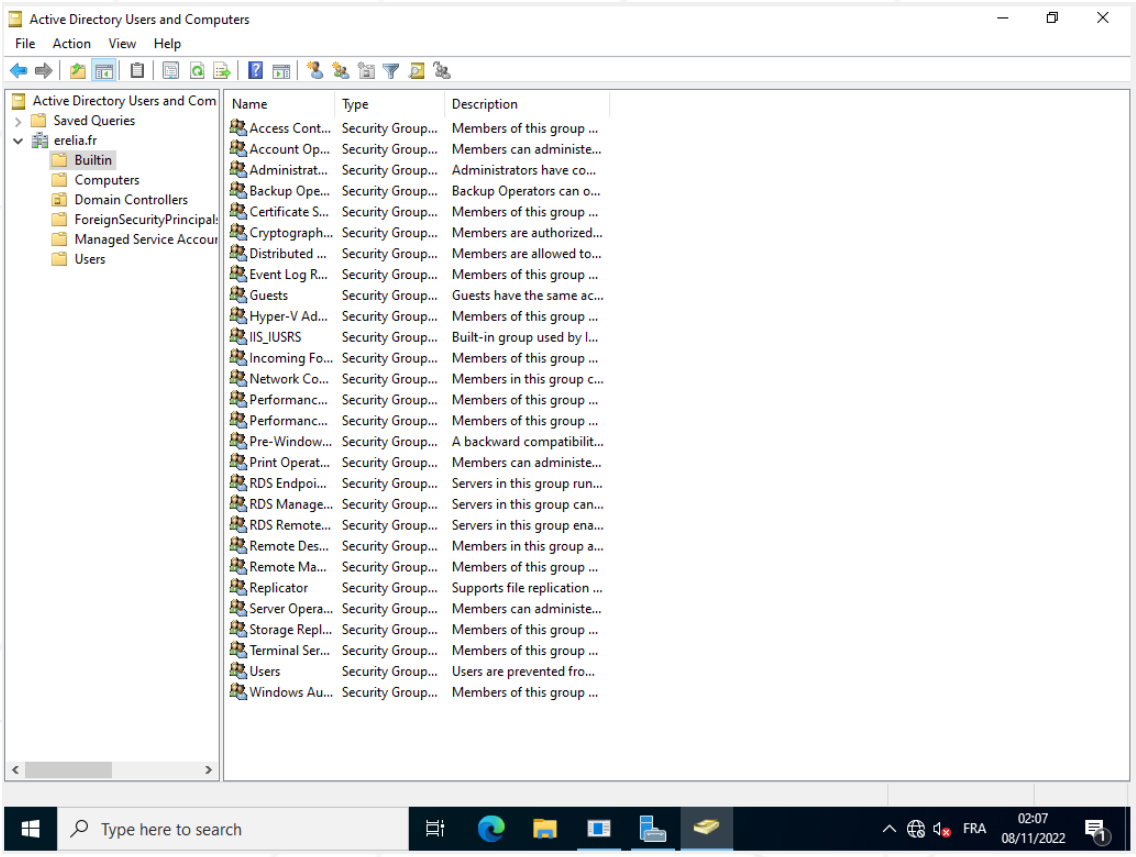
`Ensure that you give clear and descriptive names to these resources so that you can easily identify the contents they hold.`

IV.5 User account creation

In this part, you need to create a user account on each of the previously created controllers.

The account on the administration server should have the ability to view and edit the administration resources folder and the generic resources folder.

On the workshop server, the account should have the capability to view and edit the working resources folder and the generic resources folder.



You should have a screen that resembles the image above while working on this task.

Chapter V

Submission and peer-evaluation

You're provided with a helper pdf, foundable inside the attachments of the subject.
It will show you how to access the VMs created for you to complete this subject, how to lock your VMs, etc.

Once you've finish your assignment, you may lock your virtual machine, following the "Submission" section of the helper PDF. Note that, once locked, you will not be able to edit your virtual machines anymore, and a snapshot will be taken of your virtual machines at this moment, and will be reset to said snapshot before each evaluation.

So, be sure to have completly finished your work, as no modification non-requested by the scale will be acceptable during the evaluation !